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CODE A

Mock JEE Main – 6 (CBT) | JEE - 2024

Date: 29/02/2024 **Maximum Marks: 300**

Timing: 3:30 PM to 6:30 PM **Duration: 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You will NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

8. A wire of length 15 cm and radius $\sqrt{7}\times10^{-4}m$ is connected across the right gap of a meter bridge. When a resistance of 4.5Ω is connected on the left gap by using resistance box, the balance length is found to be 30 cm from the left end. If the resistivity of the wire is $R\times10^{-7}\Omega m$, then value of R is:

(A) 110 **(B)** 77 **(C)** 154 **(D)** 165

9. A convex lens of focal length 40 cm forms real image of an object. Another convex lens of focal length 20 cm forms real image of the same object for the same object distance but the image formed in this case is 36 cm closer to the lens than the previous one. What is the object distance?

(A) 60 cm **(B)** 120 cm **(C)** 30 cm **(D)** 15 cm

10. **Statement I:** Angle of contact between glass and water interface is acute.
Statement II: Water molecules are more attracted by glass molecules than by themselves.

(A) Both statement I and statement II are incorrect.
(B) Both statement I and statement II are correct.
(C) Statement I is correct but statement I is incorrect.
(D) Statement I is incorrect but statement II is correct.

11. 2 wires of same metal are connected in series. They have their length in the ratio 4 : 1 and their cross-section area in the ratio 9 : 4. The resistance of thinner wire is 20Ω. The total resistance of the combination will be:

(A) $\frac{500}{9}\Omega$ **(B)** $\frac{100}{3}\Omega$ **(C)** $\frac{81}{200}\Omega$ **(D)** $\frac{9}{200}\Omega$

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5. Given below are two statements:

Statement I: Plank's constant and Boltzmann's constant have same dimensions.

Statement II: Reynold's number is a dimensionless quantity.

In the light of above statements, choose the correct answer from the statements given below:

(A) Both Statement I and Statement II are true.

(B) Statement I is false but Statement II is true.

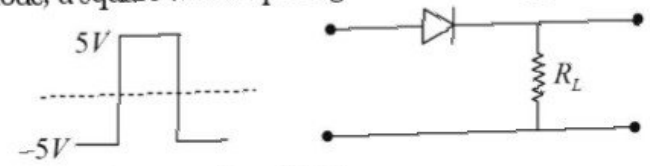
(C) Both Statement I and Statement II are false.

(D) Statement I is true but Statement II is false.

6. A body of mass 500 kg is moving horizontally with a velocity of 6 m/s. If 100 kg extra mass is added, the change in kinetic energy of body (in J) will be:

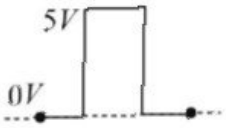
(A) 300 **(B)** –1500 **(C)** 1500 **(D)** –500

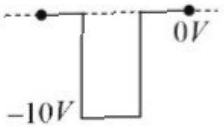
7. If in a p-n junction diode, a square wave input signal of 10V is applied as shown in figure,

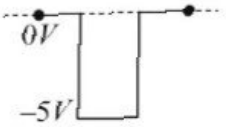


Then the value of output signal across R_L will be:

(A)


(B)


(C)


(D)


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PART I: PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The electric field of an electromagnetic wave is given by

$$7.2\times10^{3}\cos[1.5\times10^{7}z+4.5\times10^{15}t](3\hat{i}+4\hat{j})\frac{V}{m}.$$

The associated magnetic field will be:

(A) $2.4\times10^{-5}\cos(1.5\times10^{7}z+4.5\times10^{15}t)(4\hat{i}+3\hat{j})\,Wb/m^{2}$

(B) $2.4\times10^{-5}\cos(1.5\times10^{7}z+4.5\times10^{15}t)(3\hat{i}+4\hat{j})\,Wb/m^{2}$

(C) $2.4\times10^{-5}\cos(1.5\times10^{7}z+4.5\times10^{15}t)(4\hat{i}-3\hat{j})\,Wb/m^{2}$

(D) $2.4\times10^{-5}\cos(1.5\times10^{7}z+4.5\times10^{15}t)(-4\hat{i}+3\hat{j})\,Wb/m^{2}$

2. An energy of 68 eV is required to excite a hypothetical hydrogen-like atom from its 2nd Bohr orbit to 3rd. If the nuclear charge is Ze then kinetic energy of electron in 1st orbit is:

(A) $490eV$ **(B)** $520eV$ **(C)** $620eV$ **(D)** $310eV$

3. The minimum heat energy required for complete dissociation of 2 moles of H_2 gas at 300K at constant volume is:

(A) 1 kJ **(B)** 1.75 kJ **(C)** 2.5 kJ **(D)** 3.5 kJ

4. The height y and distance x along the horizontal of a projectile on a certain planet vary with time t as $y=(16t-5t^{2})m$ and $x=9t\,m$, where t is in seconds.

If range of the projectile is R and time of flight is T, then which of the following option is correct?

(A) R = 28.8 m, T = 3.2s **(B)** R = 32 m, T = 6.4s

(C) R = 14.4 m, T = 3.2s **(D)** R = 15.6 m, T = 1.6s

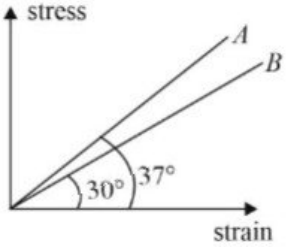
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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question should be **rounded off to the nearest integer**.

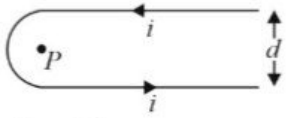
1. The stress vs strain graph for wires of materials A and B are as shown in the figure. If Y_A and Y_B are the young's modulus of the materials, then $\frac{Y_A}{Y_B}=\frac{x\sqrt{3}}{4}$. Find x.



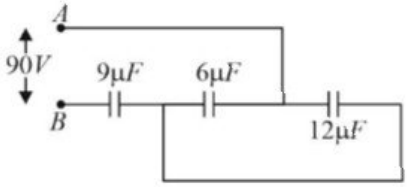
2. A particle starts motion from origin at time $t = 0$ with initial velocity $\vec{v}=4m/s\,\hat{i}$ under a constant acceleration $\vec{a}=(-2\hat{i}+3\hat{j})m/s^2$. The displacement of particle when particle starts moving parallel to y axis is $2\sqrt{x}m$. Then x is _____.

3. The curved portion as shown in the figure is a semicircle and the straight wires are long P is the centre of curved portion of the wire. The magnetic field at P is $\frac{12x}{7}\mu T$. $(i=1A \;\&\; d=5cm)$.

Then x = _____.



4. The charge on the $6\mu F$ capacitor is (in μC) _____.



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16. 2 bodies A and B moving towards each other of mass 5 kg and 2 kg respectively and speed 4 m/s & 2 m/s respectively collide head on with each other. If coefficient of restitution $e = \frac{1}{2}$. The final speed(s) (in m/s) for A and B are respectively.

(A) $\frac{10}{7}, 14$ **(B)** $\frac{31}{7}, \frac{5}{7}$ **(C)** $\frac{10}{7}, \frac{31}{7}$ **(D)** $\frac{5}{7}, \frac{10}{7}$

17. A car is travelling at 57.6 km/h on an unbanked (horizontal) circular road of radius $\frac{160}{3} m$. If the coefficient of friction between the road and the car is 0.8, then the maximum tangential deceleration that the driver of car can achieve by applying the brakes at this moment is:

(A) $8 m/s^2$ **(B)** $4.8 m/s^2$ **(C)** $6.4 m/s^2$ **(D)** $16\ m/s^2$

18. A particle of charge q and velocity v passes undeflected through a space with non-zero electric field E and magnetic field B. The undeflecting condition will hold if:

(A) $\vec{E}$ and $\vec{B}$ are parallel **(B)** $\vec{E}$ and $\vec{B}$ are antiparallel

(C) $(\vec{v} \times \vec{B})$ is parallel to $\vec{E}$ **(D)** $(\vec{B} \times \vec{v})$ is parallel to $\vec{E}$

19. One centimeter on the main scale of vernier caliper is divided into ten equal parts. If 10 divisions of vernier scale coincide with 8 small divisions of the main scale, the least count of the caliper is:

(A) 0.01 cm **(B)** 0.02 cm **(C)** 0.05 cm **(D)** 0.07 cm

20. The ratio of rms speed of molecules in Nitrogen gas to that in Helium gas at 300 K is:

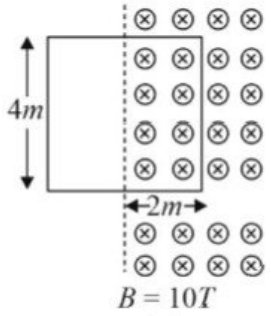
(A) $\sqrt{\frac{2}{7}}$ **(B)** $\sqrt{\frac{1}{7}}$ **(C)** $\sqrt{\frac{3}{5}}$ **(D)** $\sqrt{\frac{6}{5}}$

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12. The refracting angle of a prism for a monochromatic light is 74° and refractive index is $\frac{4}{3}$. For minimum deviation angle of incidence will be :

(A) 37° **(B)** 30° **(C)** 53° **(D)** 60°

13. A square loop of side 4 m is placed in a region of magnetic field as shown. The magnetic field is switched off in 5s. What is the charge flown through it (in C) if the resistance per unit length is $1\Omega/m$?



(A) 2C **(B)** 1C **(C)** 4C **(D)** 5C

14. The mass of a planet is eighty one times mass of earth and radius is half that of earth. If time period of a simple pendulum on surface of earth is 4s, then its time period on the surface of planet is (in SI):

(A) $\frac{1}{4}$s **(B)** 3s **(C)** $\frac{2}{9}s$ **(D)** 4s

15. 3 charges q, $2q$, $-3q$ are located at the vertices of an equilateral triangle. At the centre of the triangle:

(A) The field is zero but potential is non-zero.

(B) The field is non-zero but potential is zero.

(C) Both field and potential are zero.

(D) Both field and potential are non-zero.

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PART II: CHEMISTRY	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. During qualitative analysis of salt having cation Y^{2+}, on addition of reagent (X) a yellow precipitate (Z) is formed. Then cation Y^{2+}, reagent X & precipitate Z are?

(A) Pb^{2+}, KI, PbI_2 **(B)** Cu^{2+}, KI, Cu_2I_2

(C) Ni^{2+}, H_2S, NiS **(D)** Zn^{2+}, H_2S, ZnS

2. Consider the following complex ions:

$P \rightarrow [Fe(CN)_6]^{4-}$ $Q \rightarrow [Fe(H_2O)_6]^{2+}$ $R \rightarrow [Cr(NH_3)_6]^{3+}$

The correct order of the complex ions, according to their spin only magnetic moment values (in BM) is:

(A) P > Q > R **(B)** Q > P > R **(C)** Q > R > P **(D)** R > P > Q

3. Electronic configuration of Ytterbium in its most common oxidation state is:

(A) $[Xe]4f^{14}6s^2$ **(B)** $[Xe]4f^{13}$ **(C)** $[Xe]4f^{14}$ **(D)** $[Xe]4f^{14}5d^1$

4. Given below are two statements.

Statements I: Degree of hydrolysis & pH of a salt, NH_4CN is independent of concentration of NH_4CN.

Statements II: The solution of NH_4CN in water has pH slightly greater than 7.

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

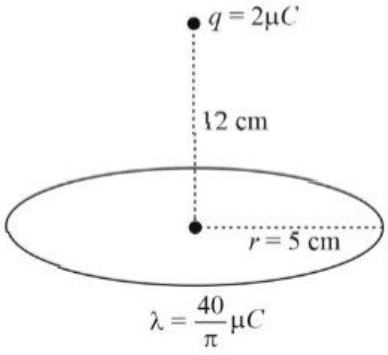
(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

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8. An electric charge $q = 2\mu C$ is fixed on the axis of a uniformly charged ring of radius 5 cm lying on a horizontal floor. The ring is in equilibrium and has a charge density $\lambda = \frac{40}{\pi}\mu C/m$. Mass of the ring is $\frac{826}{(13)^3}$ kg. The normal reaction acting on the ring is $\frac{x}{13}N$. Then x is _____.

(Take $g = 10 m/s^2$).



9. A particle executes simple harmonic motion with an amplitude of 20 cm. At mean position, velocity of the particle is 10 cm/s. Speed of the particle when it has covered phase equal to $\frac{\pi}{6}$ radians w.r.t mean position is $\frac{\sqrt{3}}{x} m/s$. Then x is _____.

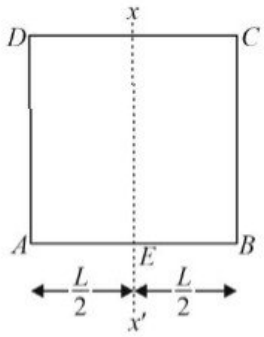
10. The energy required to separate a nucleus ${}_{50}Sn^{120}$ into its constituent nucleus is $203x$. If atomic masses of Sn^{120}, H^1 & ${}_0n^1$ are $119.9022u$, $1.0078u$ and $1.0086u$ respectively then value of x to closest integer is: $\left(1u = 931\frac{MeV}{c^2}\right)$

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5. A biconvex thick lens is constructed with glass $(\mu = 1{\cdot}5)$. Each of the surface has a radius of curvature of 20 cm and the thickness of the middle is 4cm. The image of an object placed far away from the lens is $\frac{560}{x}cm$ from the farther surface on the other side of lens. Then x = _____.

6. A boy pedals a stationary bicycle at $\frac{1}{\pi}$ revolution per second. The pedals are attached to 100 turns of coil of area $0.1m^2$ and placed in a uniform magnetic field of 0.1T. The maximum voltage generated in the coil (in V) is _____.

7. Four thin metal rods each of mass 'M' and length L are welded to form a square $ABCD$ as shown in the figure. The moment of inertia of the composite structure about a line which bisects rod AB and CD is $\frac{2ML^2}{x}$. Then x is _____.



SPACE FOR ROUGH WORK

12. Which of the following presents the correct order of the acidity in the given compounds?

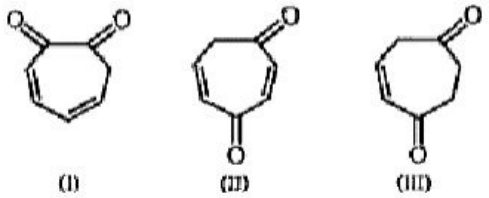
(A) $BrCH_2COOH > ClCH_2COOH > FCH_2COOH > CH_3COOH$

(B) $FCH_2COOH > ClCH_2COOH > BrCH_2COOH > CH_3COOH$

(C) $CH_3COOH > BrCH_2COOH > ClCH_2COOH > FCH_2COOH$

(D) $FCH_2COOH > CH_3COOH > BrCH_2COOH > ClCH_2COOH$

13.



(I) (II) (III)

Among the given compounds, the correct order of enol content is:

(A) I > II > III **(B)** III > II > I **(C)** II > I > III **(D)** II > III > I

14. Tropone  is _____ type of an organic compound.

(A) Benzenoid aromatic **(B)** Alicyclic

(C) Non-benzenoid non-aromatic **(D)** Non-benzenoid aromatic.

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9. Insulin has 51 amino acids in two polypeptide chains, these polypeptide chains are cross-linked by:

(A) Peroxy linkage **(B)** Disulphide linkage

(C) Glycosidic linkage **(D)** Peptide linkage

10. Given below are two statements one is labelled as Assertion **(A)** & the other is labelled as Reason (R).

Assertion (A): Boron only forms covalent compounds.

Reason (R): Small size and very high value of sum of first three ionization enthalpies of boron prevent it to form + 3 ions.

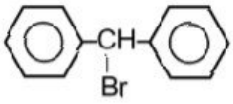
(A) A is false but R is true

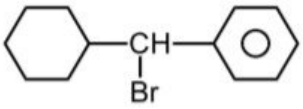
(B) Both A and R are correct and R is the correct explanation of A

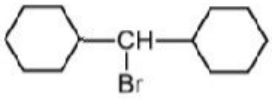
(C) A is true but R is false

(D) Both A and R are correct and R is not the correct explanation of A

11. The rate of S_N1 reaction is fastest with:

(A) 

(B) 

(C) 

(D) 

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5. Given below are two statements.

Statements I: Ortho-nitrophenol is less soluble in water than the meta & para isomers.

Statements II: Meta & Para nitrophenols exist as associated molecules.

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are incorrect

(C) Both Statement I and Statement II are correct

(D) Statement I is correct but Statement II is incorrect

6. Given below are two statements.

Statements I: Helium & Beryllium are chemically inert due to similar outer electronic configuration.

Statements II: The first ionisation enthalpy of Aluminium is lower than that of Magnesium.

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are incorrect

(C) Both Statement I and Statement II are correct

(D) Statement I is correct but Statement II is incorrect

7. Which of the following molecules are expected to exhibit intermolecular H-bonding?

(I) Acetic acid (II) o-nitrophenol (III) m-nitrophenol (IV) ortho-boric acid

Select correct alternative:

(A) I, II, III **(B)** I, II, IV **(C)** I, III, IV **(D)** II, III, IV

8. Which of the following ions has spin only magnetic moment equal to 2.83 BM?

(Ti = 22, Cr = 24, Mn = 25, Ni = 28)

(A) Ti^{3+} **(B)** Ni^{2+} **(C)** Cr^{3+} **(D)** Mn^{2+}

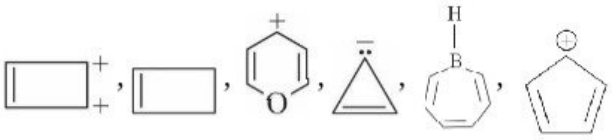
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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question should be **rounded off to the nearest integer**.

1. Among the given organic compounds, the total number of aromatic compounds are:



2. W is the weight of iron (in gram) which will be converted into Fe_3O_4 by the action of 18 gram of steam. The value of W/7 is _____. (Molar weight of Fe = 56 g/mol)

$(Fe + H_2O \rightarrow Fe_3O_4 + H_2)$

3. During the electrolysis of a concentrated brine solution, the moles of chlorine gas produced by the passage of 4F electricity is_____.

4. 3, 4-dimethylhex-3-ene on reaction with HBr in pressence of peroxide forms an addition product (A). The number of possible stereoisomers for A is _____.

5. The number of electrons in argon atom having $m_l = 1$ are _____.

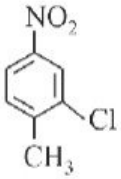
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18. A metal salt solution forms a yellow precipitate (A) with potassium chromate in acetic acid, a white precipitate (B) with dilute sulphuric acid, but gives no precipitate with sodium chloride or iodide. Then A & B are respectively:

(A) $BaCrO_4$, $Ba(NO_3)_2$ **(B)** $Ba(NO_3)_2$, $BaSO_4$

(C) $BaCrO_4$, $BaSO_4$ **(D)** $BaSO_4$, $BaCrO_4$

19. IUPAC name of the following compound is:



(A) 4-methyl-5-chloro-nitrobenzene **(B)** 2-chloro-1-methyl-4-nitrobenzene

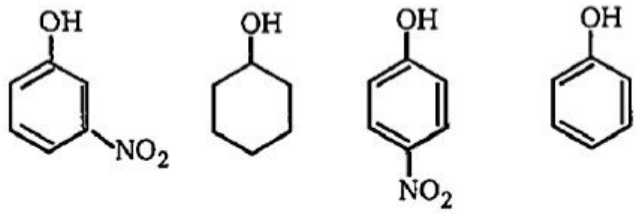
(C) 2-chloro-4-nitro-1-methylbenzene **(D)** 5-chloro-4-methylnitrobenzene

20. Which one of the following elements shows both positive & negative oxidation states?

(A) Cesium **(B)** Fluorine **(C)** Iodine **(D)** Xenon

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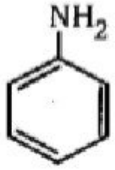
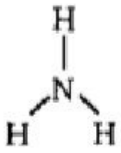
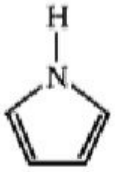
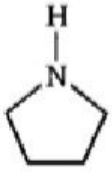
15. Rank the following compounds in order of their decreasing acidic strength (most acidic → least acidic).



(1) (2) (3) (4)

(A) 2 > 4 > 1 > 3 **(B)** 1 > 3 > 4 > 2 **(C)** 3 > 1 > 2 > 4 **(D)** 3 > 1 > 4 > 2

16. Which of the following is the strongest Bronsted base?

(A)  **(B)**  **(C)**  **(D)** 

17. If mole fraction of the solvent in a solution decreases then:

(A) Vapour pressure of solution increases **(B)** Boiling point decreases

(C) Osmotic pressure increases **(D)** All are correct.

SPACE FOR ROUGH WORK

5. Consider two straight lines and a family of straight lines $L_1 : x + y = 0$ & $L_2 : 3x - 4y = 7$ and a family of straight lines $L : y = 3x + c$, where c is parameter. The locus of mid point of segment intercepted on L between L_1 & L_2 is:

(A) $4x - 25y = 29$ **(B)** $x - y = 2$

(C) $3x - 25y = 28$ **(D)** $7x - y = 8$

6. If standard deviation of $x_i, i = 1, 2, ..., n$ is σ_X, then the standard deviation of $\frac{ax_i + b}{p}$

$\forall\ a, b, p \in R, i = 1, 2, 3, ..., n$, is:

(A) $\left|\frac{a}{p}\right|\sigma_X$ **(B)** $\left|\frac{p}{a}\right|\sigma_X$ **(C)** $\frac{p}{a}\sigma_X$ **(D)** $\frac{p}{2a}\sigma_X$

7. ${}^{35}C_0 \cdot {}^{35}C_{15} - {}^{35}C_1 \cdot {}^{35}C_{16} + {}^{35}C_2 \cdot {}^{35}C_{17} - .... + {}^{35}C_{20} \cdot {}^{35}C_{35}$ is equal to:

(A) Zero **(B)** $-{}^{35}C_{25}$ **(C)** ${}^{35}C_{25}$ **(D)** 1

8. If a, b, c are three distinct numbers in AP and $b - a, c - b, a$ are in GP, then $a : b : c =$

(A) 2 : 3 : 1 **(B)** $\frac{1}{2} : \frac{1}{3} : 1$ **(C)** 1 : 2 : 3 **(D)** 2 : 1 : 6

SPACE FOR ROUGH WORK

PART III: MATHEMATICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let $f(x)=\begin{cases}(1+ax)^{1/x} & x<0\\ b & x=0\\ \dfrac{(x+1)^{1/3}-1}{(x+1)^{1/2}-1} & x>0\end{cases}$, is continuous at $x=0$, then $3(e^a+b)$ is equal to:

(A) 3 **(B)** 6 **(C)** 4 **(D)** 8

2. Equation of the perpendicular line from $(3,-1,11)$ to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is:

(A) $\frac{x-3}{1}=\frac{y+1}{-6}=\frac{z-11}{4}$ **(B)** $\frac{x-3}{2}=\frac{y+1}{5}=\frac{z-11}{7}$

(C) $\frac{x-3}{1}=\frac{y+1}{11}=\frac{z-11}{3}$ **(D)** $\frac{x-3}{1}=\frac{y+1}{6}=\frac{z-11}{4}$

3. The radius of largest circle inscribed in the parabola, $y^2=4x$ which passes through its focus and is contained in it, is:

(A) 2 **(B)** 4 **(C)** 6 **(D)** 8

4. If x_1, y_1 are the roots of $x^2+8x-20=0$, x_2, y_2 are the roots of $4x^2+32x-57=0$ and x_3, y_3 are the roots of $9x^2+72x-112=0$, then the points, $(x_1,y_1),(x_2,y_2)$ and (x_3,y_3):

(A) Are collinear **(B)** Form an equilateral triangle

(C) Form a right angled isosceles triangle **(D)** Are concyclic

SPACE FOR ROUGH WORK

6. From the given list, the number of compounds with +6 oxidation state of sulphur are _____.

$SO_3, H_2S_2O_7, H_2S_2O_8, SO_2Cl_2, Na_2SO_3, SOF_2$

7. Sum of bond order of N_2^+ and O_2^+ is _____.

8. Consider the following data for the following reaction.

$A + B \longrightarrow AB$

S. No.	Concentration of A (in mol/L)	Concentration of B (in mol/L)	Rate (in mol L^{-1} sec^{-1})
1.	1×10^{-2}	2×10^{-2}	2×10^{-4}
2.	2×10^{-2}	2×10^{-2}	4×10^{-4}
3.	2×10^{-2}	4×10^{-2}	8×10^{-4}

The order of reaction is_____.

9. Among the following, total number of ortho and para directing groups are _____.

$-OC_2H_5, -CHO, -SO_3H, -NHCH_3, -OH, -Br, -NO_2, -CH_3$

10. An ideal gas expands against a constant external pressure of 2 atmosphere from 20 litre to 40 litre and absorbs 10.04 kJ of heat from surroundings. The change in internal energy of the system (in kJ) is _____. (Given 1 L-atm = 101 J)

SPACE FOR ROUGH WORK

18. Equation of circle passing through point of intersection of $7x+2y-9=0$ and $x-7y+6=0$ and having centre at (2, 1) is:

(A) $x^2+y^2-4x-2y+4=0$ **(B)** $x^2+y^2-2x-4y-1=0$

(C) $x^2+y^2-2x-2y+1=0$ **(C)** $x^2+y^2-2x-2y-1=0$

19. The general solution of the differential equation $\frac{dy}{dx}+\sin\frac{x+y}{2}=\sin\frac{x-y}{2}$ is:

(A) $\ell n\tan\frac{y}{2}=c-2\sin x$ **(B)** $\ell n\left(\tan\frac{y}{4}\right)=c-2\sin\left(\frac{x}{2}\right)$

(C) $\ell n\tan\left(\frac{y}{2}+\frac{\pi}{4}\right)=c-2\sin x$ **(D)** $\ell n\tan\left(\frac{y}{4}+\frac{\pi}{4}\right)=c-2\sin\left(\frac{x}{2}\right)$

20. If $|z-1|=\sqrt{5}-|z-4\sin\theta|$, $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ represents an ellipse, then exhaustive set of values of θ belongs to:

(A) $\left(-\frac{\pi}{10},\frac{3\pi}{10}\right)$ **(B)** $\left(-\frac{\pi}{5},\frac{3\pi}{10}\right)$ **(C)** $\left(-\frac{\pi}{3},\frac{\pi}{3}\right)$ **(D)** $\left(-\frac{\pi}{18},\frac{\pi}{18}\right)$

SPACE FOR ROUGH WORK

14. Let $A=\begin{bmatrix} x+\lambda & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda \end{bmatrix}$, then A^{-1} exists if:

(A) $x \neq 0$ **(B)** $\lambda \neq 0$

(C) $3x+\lambda \neq 0,\ \lambda \neq 0$ **(D)** $x \neq 0,\ \lambda \neq 0$

15. A mapping $f: R \to R$ defined by $f(x)=(x-\alpha)(x-\beta)(x-\gamma)$, where $\alpha, \beta, \gamma \in R$, $\alpha \neq \beta \neq \gamma$. The mapping is:

(A) One-to-one but not onto **(B)** Onto but not one-one

(C) Both one-one and onto **(D)** Neither one-one nor onto

16. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors such that $\left|\vec{a}+2\vec{b}+3\vec{c}\right|=\sqrt{3+2\sqrt{2}}$, angle between $\vec{a}$ and $\vec{b}$ is α, angle between $\vec{a}$ and $\vec{c}$ is β and angle between $\vec{b}$ and $\vec{c}$ is θ such that $\theta \in \left[\frac{\pi}{2}, \frac{2\pi}{3}\right]$, then the maximum value of $4\cos\alpha + 6\cos\beta$ is equal to:

(A) $2\sqrt{2}+5$ **(B)** $5-2\sqrt{2}$ **(C)** $2\sqrt{2}-5$ **(D)** $4\sqrt{2}$

17. The sum of the coefficients of all the even powers of x in the expansion of $(2x^2-3x+1)^{11}$ is:

(A) $2 \cdot 6^{10}$ **(B)** $3 \cdot 6^{10}$ **(C)** 6^{11} **(D)** 6^9

SPACE FOR ROUGH WORK

9. Let P be a variable point on the ellipse $3x^2+4y^2=12$ whose foci are F_1 and F_2. The incentre of the triangle PF_1F_2 always lies on a conic whose eccentricity is e. Then $81e^2$ is equal to:

(A) 54 **(B)** 45 **(C)** 27 **(D)** 72

10. Shortest distance between the lines $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{1}$ is equal to:

(A) $\sqrt{14}$ **(B)** $\sqrt{7}$ **(C)** $\sqrt{2}$ **(D)** $\sqrt{3}$

11. Let $I-\{0\}$ be the set of all non-zero integers and a relation R is defined as $(x,y)\in R \Rightarrow x^y=y^x$ for $x,y\in I-\{0\}$, then relation R is:

(A) Reflexive but not symmetric **(B)** Symmetric but not reflexive

(C) Reflexive and symmetric both **(D)** Neither reflexive nor symmetric

12. The value of $\lim\limits_{x\to 0}\frac{x+2\sin x}{\sqrt{x^2+2\sin x+1}-\sqrt{\sin^2 x-x+1}}$ is:

(A) Zero **(B)** 3 **(C)** 2 **(D)** 1

13. $\int\limits_{-1}^{1}\frac{dx}{(1+e^x)(1+x^2)}=$

(A) $\frac{\pi}{2}$ **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{8}$ **(D)** $\frac{\pi}{16}$

SPACE FOR ROUGH WORK

Solutions to Mock JEE MAIN – 6 (CBT) | JEE 2024

PHYSICS

SECTION-1

1.(C) $E_0 = 7.2\times5\times10^3\ V/m; \qquad C = \dfrac{4.5\times10^{15}}{1.5\times10^7} = 3\times10^8 m/s.$

$B_0 = \dfrac{E_0}{C} = \dfrac{7.2\times5\times10^3}{3\times10^8} = 2.4\times5\times10^{-5} Wb/m^2$

$\hat{B} = \hat{c}\times\hat{E}$ Now $\hat{B} = -\hat{k}\times\dfrac{3\hat{i}+4\hat{j}}{\sqrt{25}} \Rightarrow \hat{B} = \dfrac{(4\hat{i}-3\hat{j})}{5}$

$\vec{B} = B_0\cos(1.5\times10^7 z + 4.5\times10^{15}t)\times\hat{B}$

$\Rightarrow \qquad \vec{B} = 2.4\times10^{-5}\cos(1.5\times10^7 z + 4.5\times10^{15}t)(4\hat{i}-3\hat{j})\,Wb/m^2$

2.(A) $13.6z^2\left[\dfrac{1}{4}-\dfrac{1}{9}\right] = 68eV$

$13.6z^2\left[\dfrac{5}{36}\right] = 68eV \Rightarrow z^2 = 36 \Rightarrow z = 6$

$K.E. = \dfrac{13.6Z^2}{1^2} = \dfrac{68\times36}{5} = 489.6eV$

3.(C) $U_f = 2n\left(\dfrac{3R}{2}T\right) = 3nRT; \qquad U_i = n\left(\dfrac{5R}{2}T\right) = \dfrac{5nRT}{2}$

Heat energy required is $\Rightarrow \qquad \Delta U = 3nRT - \dfrac{5}{2}nRT = \dfrac{nRT}{2}$

$\Rightarrow \qquad \Delta U = \dfrac{2}{2}\times\dfrac{25}{3}\times300 \qquad \Rightarrow \qquad \Delta U = 2500J = 2.5\ kJ$

4.(A) $y = 0 \Rightarrow 16t - 5t^2 \Rightarrow 5t^2 = 16t \Rightarrow t = 0,\ t = \dfrac{16}{5}s$

$T = 3.2\ s; \qquad R = 9T = 9\times3.2 = 28.8\ m$

5.(B)

6.(B) From momentum conservation,

$m_1v_1 = (m_1+m_2)v_2 \qquad \Rightarrow \qquad 500\times6 = (600)\times v_2 \qquad \Rightarrow \qquad v_2 = 5m/s$

$K.E._i = \dfrac{1}{2}m_1v_1^2 = \dfrac{1}{2}500\times(6)^2 = (250\times36)J$

$K.E._f = \dfrac{1}{2}(m_1+m_2)v_2^2 = \dfrac{1}{2}(600)(5)^2 = (25\times300)J$

$\Delta K.E. = 25\times300 - 250\times36$ $= 25(300-360) = -25\times60 = -1500\,J$

7.(B) When input = $-5V$. Diode is R.B. and does not conduct Hence output = 0.
When input = $5V$, diode is F.B. and conducts. Hence output = $5V$.

8.(C) $\dfrac{4.5}{R} = \dfrac{30}{70}$ by balanced wheatstone bridge condition

6. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}|=1, |\vec{b}|=4, \ \vec{a}\cdot\vec{b}=2$. If $\vec{c}=(2\vec{a}\times\vec{b})-3\vec{b}$ then the angle between $\vec{b}$ and $\vec{c}$ is θ then the value of $[\theta]$ is _____.

Where [.] is greatest integer function

7. If $y(x)$ is a solution of $\frac{dy}{dx}-\frac{xy}{1+x}=\frac{1}{1+x}$ and $y(0)=-1$, then the value of $|3y(2)|$ is _____.

8. Let $g(x)=f\left(\frac{x}{f(x)}\right)$ where $f(x)$ is a differentiable positive function on $(0,\infty)$ such that $f(1)=f'(1)$. then $g'(1)$ = _____.

9. Let $A=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$ and matrix B is defined such that $B=A+4A^2+6A^3+4A^4+A^5$. If $\det(B)=1$, then the number of values of α in $[-2\pi, 2\pi]$ is _____.

10. The value of $\cos\frac{\pi}{31}\cos\frac{2\pi}{31}\cos\frac{4\pi}{31}\cos\frac{8\pi}{31}\cos\frac{16\pi}{31}$ is equal to N, then $\left|\frac{1}{N}\right|$ equals _____.

SPACE FOR ROUGH WORK

End of 2024 | Mock JEE Main - 6 (CBT)

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question should be **rounded off to the nearest integer**.

1. Product $3^{\frac{1}{4}} \cdot 9^{\frac{1}{8}} \cdot 27^{\frac{1}{16}} \cdot 81^{\frac{1}{32}} \ldots \infty$ is equal to _____.

2. If $x^2 + x + 1 = 0$, then the value of $\frac{1}{9} \times \left[\left(x + \frac{1}{x} \right)^2 + \left(x^2 + \frac{1}{x^2} \right)^2 + \ldots + \left(x^{27} + \frac{1}{x^{27}} \right)^2 \right]$ is _____.

3. A box containing 24 balls out of which 8 are black and 16 are white. The balls are drawn at random from the box one at a time with replacement. If the probability that a white ball is drawn for the 4th time on the 7th draw is $\left(\frac{a}{b} \right)^3 \times \frac{40}{81}$, where a and b are relatively prime numbers, then the value of $(a + b + 1)$ is _____.

4. If solution of differential equation $\frac{dy}{dx} - y = 1 - e^{-x}$ and $y(0) = \frac{-1}{2}$. Then value of $\lim_{x \to \infty} |y|$ is _____.

5. Let $f(x + y) = f(x) f(y)$ for all x and y, and $f'(0) = 3$, then absolute value of $\frac{f'(15)}{e^{45}}$ is equal to _____.

SPACE FOR ROUGH WORK

SECTION-2

1.(3) $\left(\frac{\text{Stress}}{\text{Strain}}\right)_A = Y_A = \tan 37^\circ = \frac{3}{4}$

$\left(\frac{\text{Stress}}{\text{Strain}}\right)_B = Y_B = \tan 30^\circ = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad \frac{Y_A}{Y_B} = \frac{3}{4} \times \frac{\sqrt{3}}{1} \quad \Rightarrow \quad \frac{3\sqrt{3}}{4} = \frac{Y_A}{Y_B}$

2.(13) $u_x = 4,\ a_x = -2$ particle starts moving parallel to y-axis when $V_x = 0$

$\Rightarrow \quad v_x = u_x - at \quad \Rightarrow \quad 0 = 4 - 2t \quad \Rightarrow \quad t = 2s$

$x = 4 \times 2 - \frac{1}{2}(2)(2)^2 \quad \Rightarrow \quad x = 4m$

$y = \frac{1}{2} a_y t^2 \ \Rightarrow \ y = \frac{1}{2}(3)(2)^2 \quad \Rightarrow \quad y = 6m$

Displacement $= \sqrt{4^2 + 6^2} = 2\sqrt{13}\, m \quad \Rightarrow \quad x = 13$

3.(12) $B_P = B_1 + B_2 + B_3 \quad = \frac{\mu_0 i}{2\pi d} + \frac{\mu_0 i}{2d} + \frac{\mu_0 i}{2\pi d} = \frac{\mu_0 i}{2d}\left(1 + \frac{2}{\pi}\right)$

$\Rightarrow \quad B_P = \frac{\mu_0}{10 \times 10^{-2}}\left(\frac{\pi + 2}{\pi}\right) \quad \Rightarrow \quad B_P = 10 \times \frac{4\pi \times 10^{-7}}{\pi}(\pi + 2)$

$\Rightarrow \quad B_P = 40 \times 10^{-7}\left(\frac{22}{7} + 2\right) \quad \Rightarrow \quad B_P = 4 \times 10^{-6}\left(\frac{36}{7}\right)$

$\Rightarrow \quad B_P = 144 \times \frac{10^{-6}}{7} \quad \Rightarrow \quad B_P = \frac{12x}{7} \times 10^{-6} T$

$x = 12$

4.(180) $6\mu F$ and $12\mu F$ are in parallel.

$C_{eq} = \frac{18 \times 9}{18 + 9} = \frac{18 \times 9}{27} = 6\mu F$

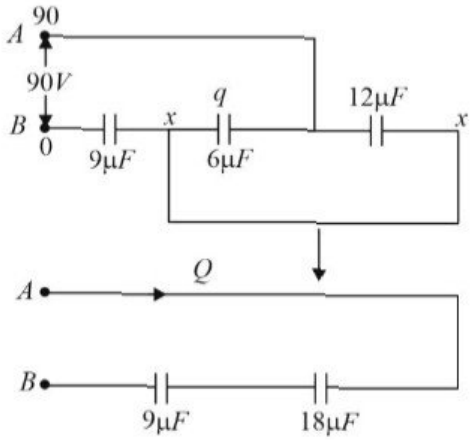
$Q = C_{eq} V = 6 \times 90 = 540\mu C$

$Q = 540\mu C$

ΔV across $18\mu F$ is $90 \times \frac{9}{27} = 30V$

So, charge across $6\mu F$ is $q = CV$

$\Rightarrow \ q = 6\mu F \times 30V = 180\mu C$



5.(29) $\frac{3}{2V_1} - \frac{1}{-\infty} = \frac{1}{2(20)}$

$\Rightarrow \quad \frac{3}{2V_1} = \frac{1}{2(20)} \ \Rightarrow \ V_1 = 60cm$

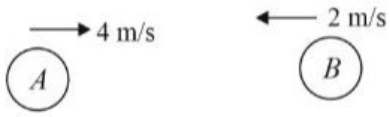
$\frac{1}{V} - \frac{3}{2(56)} = \frac{\frac{-1}{2}}{-20}$

$$T = 2\pi\sqrt{\frac{L}{g}}, \qquad T_p = 2\pi\sqrt{\frac{L}{g_p}}$$

$$T_p = 2\pi\sqrt{\frac{L}{81\times 4g}} = \frac{1}{18}2\pi\sqrt{\frac{L}{g}}$$

$$\Rightarrow \qquad T_p = \frac{1}{18}\left(2\pi\sqrt{\frac{L}{g}}\right) \qquad \Rightarrow \qquad T_p = \frac{T}{18} = \frac{4}{18}s = \frac{2}{9}s$$

15.(B)

16.(C) 

$$5\times 4 - 2\times 2 = 5V_1 + 2V_2; \qquad e = \frac{1}{2} \Rightarrow \frac{V_2 - V_1}{4-(-2)} = \frac{1}{2}$$

$$\Rightarrow \qquad V_2 - V_1 = 3 \qquad \ldots \text{(i)}$$

$$2V_2 + 5V_1 = 16 \qquad \ldots \text{(ii)}$$

$$\Rightarrow \qquad 2V_2 + 5V_1 = 16$$

$$\underline{2V_2 - 2V_1 = 6}$$

$$7V_1 = 10$$

$$\Rightarrow \qquad V_1 = \frac{10}{7}\text{m/s} \quad \text{and } V_2 = \frac{31}{7}\text{m/s}$$

17.(C) $m\sqrt{\left(\frac{V^2}{r}\right)^2 + a_t^2} = \mu mg; \qquad V = 57.6\,km/h = 16\,m/s$

$$\Rightarrow \qquad \left\{\frac{(16)^2}{160}\times 3\right\}^2 + a_t^2 = \left(\frac{4}{5}\times 10\right)^2 \qquad \Rightarrow \qquad a_t^2 = 64 - 23.04$$

$$\Rightarrow \qquad a_t = \sqrt{40.96} = 6.4\,m/s^2 \qquad \Rightarrow \qquad a_t = 6.4\,m/s^2$$

18.(D) For no deflection, $q\vec{E} + q(\vec{v}\times\vec{B}) = 0 \qquad \Rightarrow \qquad \vec{E} = -(\vec{v}\times\vec{B})$

$\Rightarrow \qquad \vec{v}\times\vec{B}$ is antiparallel to $\vec{E}$

19.(B) In this case least count = 1MSD – 1VSD

$$= 1mm - \frac{8}{10}mm = 0.02cm$$

20.(B) $\frac{1}{2}m_{N_2}v_{N_2}^2 = \frac{1}{2}m_{He}v_{He}^2$ at 300 K

$$\Rightarrow \qquad \frac{v_{N_2}}{v_{He}} = \sqrt{\frac{m_{He}}{m_{N_2}}} \qquad \Rightarrow \qquad \frac{v_{N_2}}{v_{He}} = \sqrt{\frac{4}{28}} = \sqrt{\frac{1}{7}}$$

$\Rightarrow \quad R = 4.5 \times \frac{7}{3} = \frac{31.5}{3} = 10.5\Omega \Rightarrow R = 10.5\Omega$

$\Rightarrow \quad \frac{\rho l}{A} = 10.5 \Rightarrow \frac{\rho \times 15 \times 10^{-2}}{\frac{22}{7} \times 7 \times 10^{-8}} = 10.5$

$\Rightarrow \quad \rho = \frac{10.5 \times 22}{15} \times 10^{-6} = \frac{105}{15} \times 22 \times 10^{-7} \quad \Rightarrow \quad \rho = 154 \times 10^{-7}\Omega - m$

9.(B) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{x} - \frac{1}{u} = \frac{1}{40} \quad \ldots$ (i)

$\frac{1}{x-36} - \frac{1}{u} = \frac{1}{20} \quad \ldots$ (ii)

$\Rightarrow \quad \frac{1}{x} - \frac{1}{40} = \frac{1}{x-36} - \frac{1}{20} \quad \Rightarrow \quad \frac{40-x}{40x} = \frac{20-(x-36)}{20(x-36)}$

$\Rightarrow \quad (40-x)(x-36) = 2x(56-x) \quad \Rightarrow \quad -1440 - x^2 + 36x + 40x = 112x - 2x^2$

$\Rightarrow \quad x^2 - 36x - 1440 = 0 \quad \Rightarrow \quad x = 60cm$

$\Rightarrow \quad \frac{1}{60} - \frac{1}{u} = \frac{1}{40} \Rightarrow \frac{1}{u} = \frac{1}{60} - \frac{1}{40} \quad \Rightarrow \quad \frac{1}{u} = \frac{-1}{120} \Rightarrow u = -120cm$

So, object distance is 120 cm.

10.(B)

11.(A) $\frac{R_1}{R_2} = \frac{L_1}{L_2} \times \frac{A_2}{A_1} = \frac{4}{1} \times \frac{4}{9} = \frac{16}{9}$

$\frac{R_1}{20} = \frac{16}{9} \quad \Rightarrow \quad R_1 = \frac{320}{9}\Omega$

$R_{eq} = R_1 + R_2 \quad \Rightarrow \quad 20 + \frac{320}{9} = \frac{500}{9}\Omega = R_{eq}$

12.(C) For angle of minimum deviation

$r_1 = r_2 = r = \frac{A}{2} \Rightarrow r = 37^\circ$

$\frac{\sin i}{\sin r} = u \Rightarrow \sin i = \sin r \times u$

$\Rightarrow \quad \sin i = \frac{3}{5} \times \frac{4}{3} \Rightarrow \sin i = \frac{4}{5}$

$\Rightarrow \quad i = 53^\circ$

13.(D) $\Delta Q = \frac{\Delta\phi}{R} = \frac{A(\Delta B)}{R} = \frac{8 \times 10}{16} = 5C$

$\Delta Q = 5C$

14.(C) $g = \frac{GM}{R^2}, \qquad g_p = \frac{G(81M)}{(R/2)^2} = 81 \times 4(g)$

$$\Rightarrow \quad \frac{1}{V}-\frac{1}{40}=\frac{3}{112}=\frac{14+15}{560} \quad \Rightarrow \quad \frac{1}{V}=\frac{29}{560}$$

$$\Rightarrow \quad V=\frac{560}{29}cm; \qquad x=29$$

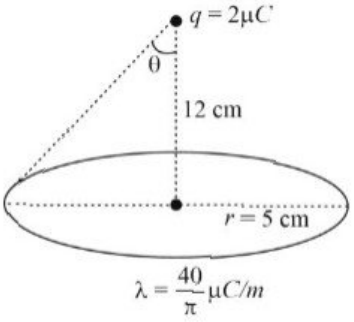
6.(2) $\varepsilon = NBA\omega \Rightarrow \quad \varepsilon = 100\times\frac{1}{10}\times\left(\frac{1}{10}\right)\times\left(\frac{1}{\pi}\right)\times 2\pi = 2V$

7.(3) $I_{xx'}=\frac{2ML^2}{12}+2M\left(\frac{L}{2}\right)^2$

$$=\frac{2ML^2}{12}+\frac{2ML^2}{4} \Rightarrow I_{xx'}=\frac{4ML^2}{6}$$

$$\Rightarrow \quad I_{xx'}=\frac{2ML^2}{3} \qquad \Rightarrow \quad x=3$$

8.(100) Force between point charge and ring is $F_e=\frac{KQx}{(R^2+x^2)^{3/2}}\times q$



$$F_e=\frac{K\times(\lambda 2\pi r)\times q\times x}{(R^2+x^2)^{3/2}}$$

$$\Rightarrow \quad F_e=\frac{9\times10^9\times\frac{40}{\pi}\times10^{-6}\times2\pi\times5\times10^{-2}\times2\times10^{-6}\times12\times10^{-2}}{(13\times10^{-2})^3}$$

For equilibrium $mg+F_e=N$

$$N=\frac{8260}{(13)^3}+\frac{8640}{(13)^3}; \qquad N=\frac{16900}{(13)^3}=\frac{100}{13}$$

9.(20) $v=\omega\sqrt{A^2-x^2}$

When the particle has covered phase equal to $\frac{\pi}{6}$ radians w.r.t. mean position,

$$x=A\sin\frac{\pi}{6} \Rightarrow x=\frac{A}{2}. \qquad \Rightarrow \quad v=\omega\sqrt{A^2-\frac{A^2}{4}}$$

$$\Rightarrow \quad v=\frac{\sqrt{3}}{2}A\omega \quad \Rightarrow \quad v=\frac{\sqrt{3}}{2}v_{max}$$

$$\Rightarrow \quad v=\frac{\sqrt{3}}{2}\times\frac{1}{10} \quad \Rightarrow \quad v=\frac{\sqrt{3}}{20}\text{m/s} \quad \Rightarrow \quad x=20$$

10.(5) The mass defect $\Delta m = Zm_H+Nm_n-M_{Sn}$

$$=(50\times1.0078+70\times1.0086-119.9022)$$
$$=(50.39+70.602-119.9022)$$
$$=1.0898u$$

Binding energy $=\Delta mc^2$

$$=1.0898\times931$$
$$=1014.6038 \simeq 203\times5MeV$$

mass of Fe required $= \frac{3}{4} \times 56$

$\frac{W}{7} = \frac{3}{4} \times \frac{56}{7} = 6$

3.(2) Anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$

moles of Cl_2 formed $= \frac{1}{2} \times 4 = 2$

4.(4)

$$CH_3 - CH_2 - \underset{\underset{CH_3}{|}}{C} = \underset{\underset{CH_3}{|}}{C} - CH_2 - CH_3 + HBr \xrightarrow{\text{peroxide}} \quad A$$

$$CH_3 - CH_2 - \overset{*}{C}H(CH_3) - \overset{*}{C}(Br)(CH_3) - CH_2 - CH_3$$

(A)

Number of chiral centres = 2

Number of stereoisomers = $2^2 = 4$

5.(4) ${}_{18}Ar = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

number of electrons having $m_l = 1$, are 4.

6.(4) Oxidation state of S in SO_3, $H_2S_2O_7$, $H_2S_2O_8$ and SO_2Cl_2 is + 6

7.(5) Bond order of $N_2^+ = 2.5$

bond order of $O_2^+ = 2.5$

sum = 5

8.(2) Let order with respect to A = x

order with respect to B = y

rate = $k[A]^x[B]^y$

$2\times10^{-4} = k(1\times10^{-2})^x(2\times10^{-2})^y$ ...(i)

$4\times10^{-4} = k(2\times10^{-2})^x(2\times10^{-2})^y$ ...(ii)

$8\times10^{-4} = k(2\times10^{-2})^x(4\times10^{-2})^y$ ...(iii)

from (i)/(ii) $\Rightarrow x = 1$

from (ii)/(iii) $\Rightarrow y = 1$

order of reaction $= x + y = 1 + 1 = 2$

9.(5) $-OC_2H_5, -NHCH_3, -OH, -Br, -CH_3$ are ortho and para directors.

10.(6)

$q = 10.04\,kJ$

$W = -P_{ext} \times (V_2 - V_1) = -2 \times (40 - 20)$

$W = -40\ L - atm$

$W = -40 \times 101\,J$

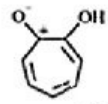
$W = -4040\ J$

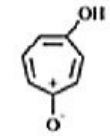
$W = -4.04\,kJ$

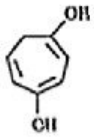
$\Delta U = q + W \Rightarrow 10.04 - 4.04 \Rightarrow \Delta U = 6\ kJ$

12.(B) More electronegative atom is attached therefore more stable anion (conjugate ion) is formed. Hence more acidic in nature.

13.(A)

(I)  Aromatic + Hydrogen bonding (due to -I effect of carbonyl. Hence more enol content will be formed)

(II)  Aromatic

(III)  Non-aromatic

14.(D)

15.(D) In 3rd compound,

–R effect of NO_2 is present.

In 1st compound, –I effect of NO_2 is present.

In 4th compound, only resonance is present.

16.(D) In option (A) and (C) lone pair is involved in resonance, hence unavailable. Secondary aliphatic amine is more basic than ammonia.

17.(C) Mole fraction of solvent decreases means mole fraction of solute increases.

18.(C) $Ba(NO_3)_2 + K_2CrO_4 \rightarrow \underset{\text{yellowppt.}}{\underset{(A)}{BaCrO_4}}\downarrow$

$Ba(NO_3)_2 \xrightarrow{\text{dil. } H_2SO_4} \underset{\text{whiteppt.}}{\underset{(B)}{BaSO_4}}\downarrow$

19.(B)

20.(C) Cesium → positive & zero

Fluorine → Negative & zero

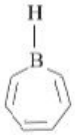
Iodine → Positive, Negative & zero

Xenon → Positive & zero

SECTION-2

1.(3)  has 2π electrons and cyclic resonance, hence aromatic.

 has 6π electrons and cyclic resonance, hence aromatic.

 has 6π electrons and cyclic resonance, hence aromatic.

2.(6) $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$

moles of $H_2O = \frac{18}{18} = 1$

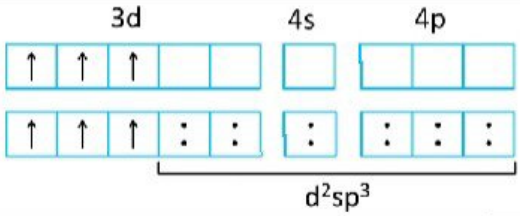
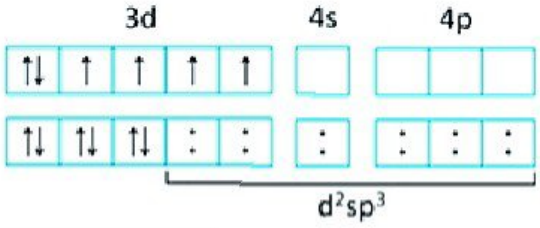
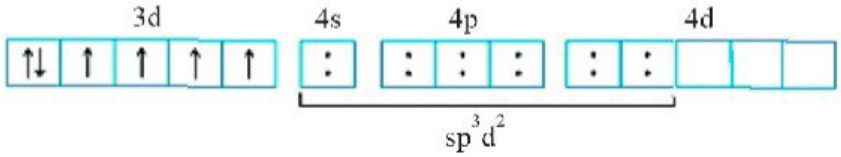
moles of Fe required $= \frac{3}{4}$

CHEMISTRY

SECTION-1

1.(A) $\underset{Y^{2+}}{Pb^{2+}} + \underset{X}{KI} \rightarrow \underset{(yellow)Z}{PbI_2}\downarrow$

2.(C) P : n = 0
Q : n = 4
R : n = 3

R →	$Cr^{3+}\left(d^3\right)$	
	$[Cr(NH_3)_6]^{3+}$	
P →	$Fe^{2+}\left(d^6\right)$	
	$\left[Fe(CN)_6\right]^{4-}$	
Q →	$\left[Fe(H_2O)_6\right]^{2+}$	

3.(B) Electronic configuration of Ytterbium = $[Xe]4f^{14}6s^2$.

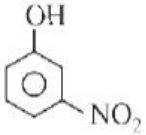
Most common oxidation state of [Lanthanoids] is +3 thus $Y^{3+} = [Xe]4f^{13}$.

4.(B) NH_4OH is relatively stronger base than acid HCN.

5.(C) Ortho nitrophenol is involved in intra molecular hydrogen bonding and meta & para isomers of nitrophenols involved in intermolecular hydrogen bonding with the other molecules.

6.(A) Beryllium is not chemically inert and form compounds easily.

7.(C) Intermolecular H-bonding occurs in:

Acetic acid	CH_3COOH
m-nitrophenol	
ortho-boric acid	H_3BO_3

8.(B) $\mu = \sqrt{n(n+2)}$

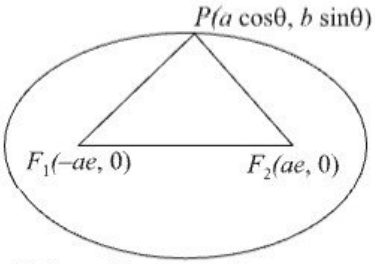
$Ni^{2+} = 3d^8; n = 2$

9.(B) Between two polypeptide chains disulphide linkage is present.

10.(B) Due to small size of boron, its ionization to +3 state is difficult, thus it forms covalent compounds.

11.(A) Most stable carbocation is formed in option **(A).**

9.(A)



$PF_1 = a(1+e\cos\theta)$

$PF_2 = a(1-e\cos\theta)$

$F_1F_2 = 2ae$

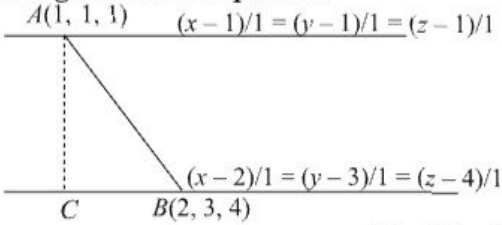
In centre (I) of $\Delta PF_1F_2 \equiv \left(ae\cos\theta, \frac{be}{1+e}\sin\theta\right)$

The locus of incentre of triangle PF_1F_2 is an ellipse whose eccentricity is $\sqrt{\frac{2e}{1+e}}$, where $e = \frac{1}{2}$ is the eccentricity of the given ellipse.

$\therefore \quad \sqrt{\frac{2e}{1+e}} = \sqrt{\frac{2}{3}}$

$\Rightarrow \quad 81e^2 = 54$

10.(C) The given lines are parallel.



From the figure, we get $BC = \frac{(2-1)1}{\sqrt{3}} + \frac{(3-1)1}{\sqrt{3}} + \frac{(4-1)1}{\sqrt{3}}$

$= \frac{1+2+3}{\sqrt{3}} = 2\sqrt{3}$ $\quad (\because BC = \left|\frac{\vec{AB}\cdot\vec{n}}{|\vec{n}|}\right|$ where $\vec{n}$ is vector parallel to line BC)

$AB = \sqrt{1+4+9} = \sqrt{14}$

Shortest distance $= AC = \sqrt{14-12} = \sqrt{2}$

11.(C) $\because \quad (x, y) \in R \Rightarrow x^y = y^x$

$\therefore \quad (x, x) \in R$ as $x^x = x^x \; \forall \; x \in I - \{0\}$

$\therefore \quad R$ is reflexive

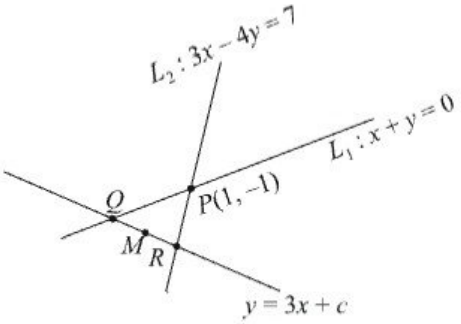
Now $\quad (x, y) \in R \Rightarrow x^y = y^x \Rightarrow y^x = x^y$

$\Rightarrow \quad (y, x) \in R \quad \therefore R$ is symmetric

12.(C) $\lim\limits_{x\to 0} \dfrac{(x+2\sin x)\left(\sqrt{x^2+2\sin x+1}+\sqrt{\sin^2 x-x+1}\right)}{x^2+2\sin x-\sin^2 x+x}$

$\lim\limits_{x\to 0} \dfrac{(1+2\frac{\sin x}{x})\left(\sqrt{x^2+2\sin x+1}+\sqrt{\sin^2 x-x+1}\right)}{x+2\frac{\sin x}{x}-\frac{\sin^2 x}{x}+1} = \frac{3\times 2}{3} = 2$

5.(C)



We get, $Q \equiv \left(-\frac{c}{4}, \frac{c}{4}\right)$, $R \equiv \left(\frac{7+4c}{-9}, \frac{7+c}{-3}\right)$

Let $M=(h,k)$

Then, $2h = \frac{-7}{9} - \frac{25}{36}c$, $2k = \frac{-7}{3} + \frac{c}{4} - \frac{c}{3}$

$\Rightarrow \quad 3h - 25k = 28$

Therefore, required locus $3x - 25y = 28$

6.(A) Let $u = \frac{ax_i + b}{p} \quad \therefore \quad \bar{u} = \frac{a\bar{x} + b}{p}$

$$\therefore \quad SD = \sqrt{\frac{\sum (u-\bar{u})^2}{\sum f}} = \sqrt{\frac{\sum \left(\frac{ax_i+b}{p} - \frac{a\bar{x}+b}{p}\right)^2}{\sum f}}$$

$$= \sqrt{\frac{\sum \left(\frac{ax_i+b-a\bar{x}-b}{p}\right)^2}{\sum f}} = \sqrt{\frac{a^2}{p^2}\frac{\sum (x_i-\bar{x})^2}{\sum f}} = \sqrt{\frac{a^2}{p^2}\sigma_x^2} = \left|\frac{a}{p}\right|\sigma_x$$

7.(C) $(x+1)^{35}(1-x)^{35} = (1-x^2)^{35}$

$\Rightarrow \quad ({}^{35}C_0 x^{35} + {}^{35}C_1 x^{34} + ... + {}^{35}C_{35})$

$({}^{35}C_0 - {}^{35}C_1 x + {}^{35}C_2 x^2 .... - {}^{35}C_{35} x^{35}) = (1-x^2)^{35}$

$\Rightarrow \quad -{}^{35}C_0 \cdot {}^{35}C_{15} + {}^{35}C_1 \cdot {}^{35}C_{16} - ..... - {}^{35}C_{20}\,{}^{35}C_{35} =$

Co-efficient of x^{50} in $(1-x^2)^{35} = -{}^{35}C_{25}$

$\therefore \quad {}^{35}C_0 \cdot {}^{35}C_{15} - {}^{35}C_1 \cdot {}^{35}C_{16} + ... + {}^{35}C_{20} \cdot {}^{35}C_{35} = {}^{35}C_{25}$

8.(C) a, b, c are in AP $\Rightarrow \quad a + c = 2b$ ....(1)

$(b-a), (c-b), a$ are in GP

$\therefore \quad (c-b)^2 = (b-a)a$

$(2b-a-b)^2 = (b-a)a$ [from (1)]

$\Rightarrow \quad (b-a)^2 = a(b-a)$

$\Rightarrow \quad \therefore b - a = a \quad (\because b \neq a)$

$b = 2a$ .....(2)

Again from (1) and (2) we get

$a + c = 4a \qquad c = 3a$

$a:b:c = a:2a:3a = 1:2:3$

MATHEMATICS

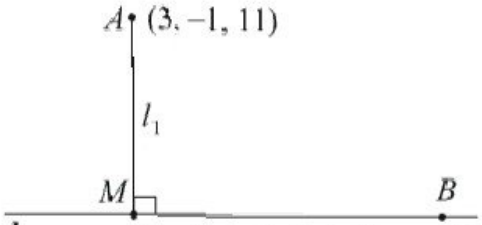
SECTION-1

1.(C) $f(0^-) = e^a$

$f(0) = b$

$f(0^+) = \frac{2}{3} \Rightarrow b = e^a = \frac{2}{3}$

2.(A)



Direction ratios of AM is $< 2\lambda - 3, 3\lambda + 3, 4\lambda - 8 >$

As $l_1 \perp l_2$

$\Rightarrow \quad 2(2\lambda - 3) + 3(3\lambda + 3) + 4(4\lambda - 8) = 0$

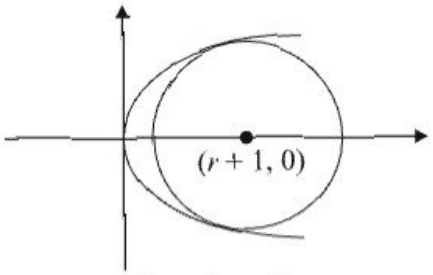
$\Rightarrow \quad \lambda(4 + 9 + 16) - 6 + 9 - 32 = 0$

$\Rightarrow \quad \lambda = 1$

$M(2, 5, 7)$

Dr's of AM are $< -1, 6, -4 >$

3.(B)



$(x - r - 1)^2 + y^2 = r^2$

$\Rightarrow \quad (x - r - 1)^2 + 4x = r^2$

$\Rightarrow \quad x^2 + (4 - (2r + 2))x + 2r + 1 = 0$

$D = 0 \Rightarrow r = 4$

4.(A) Given $x_1 + y_1 = -8 \Rightarrow y_1 = -8 - x_1;$

$x_2 + y_2 = -8 \Rightarrow y_2 = -8 - x_2$

and $x_3 + y_3 = -8 \Rightarrow y_3 = -8 - x_3$

now check

$\begin{vmatrix} x_1 & -8 - x_1 & 1 \\ x_2 & -8 - x_2 & 1 \\ x_3 & -8 - x_3 & 1 \end{vmatrix}$ value of this determinant comes out to be zero.

$\therefore$ Points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are collinear.

$$x^3+\frac{1}{x^3}=1+1=2$$

$$x^4+\frac{1}{x^4}=\left(x^2+\frac{1}{x^2}\right)^2-2=1-2=-1$$

$$x^5+\frac{1}{x^5}=\left(x^4+\frac{1}{x^4}\right)\left(x+\frac{1}{x}\right)-\left(x^3+\frac{1}{x^3}\right)$$

$$=1-2=-1$$

$$x^6+\frac{1}{x^6}=\left(x^3+\frac{1}{x^3}\right)^2-2=4-2=2$$

Hence, the value of required expression $=\frac{1}{9}(9\times 4+18)=\frac{1}{9}\times 54=6$

3.(6) p = probability of drawing a white ball $=\frac{16}{24}=\frac{2}{3}$

q = probability of drawing a black ball $=\frac{8}{24}=\frac{1}{3}$

Probability of drawing a white ball 4th time on the 7th draw.
= Probability of drawing 3 white balls in first 6 draws × probability of drawing a white ball at the 7th draw.

$$=\left({}^6C_3p^3q^3\right)\times\frac{16}{24}=\left(20\times\left(\frac{2}{3}\right)^3\left(\frac{1}{3}\right)^3\right)\times\frac{2}{3}=\left(\frac{2}{3}\right)^3\times\frac{40}{81}$$

Clearly $a=2$, $b=3$.
Hence, the value of $a+b+1=$ is $2+3+1=6$

4.(1) $\frac{dy}{dx}-y=1-e^{-x}$, $I.F.=e^{-x}$

$$\therefore \quad y\cdot e^{-x}=\int(e^{-x}-e^{-2x})\,dx$$

$$y\cdot e^{-x}=-e^{-x}+\frac{1}{2}e^{-2x}+C$$

$$y=-1+\frac{1}{2}e^{-x}+C.e^{x}$$

$$\Rightarrow \quad y(0)=-1+\frac{1}{2}+C=\frac{-1}{2}$$

$$\Rightarrow \quad C=0$$

$$\lim_{x\to\infty}|y|=\lim_{x\to\infty}\left|-1+\frac{1}{2}e^{-x}\right|=1$$

5.(3) $$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}f(x)\left(\frac{f(h)-1}{h}\right)=f(x)\cdot f'(0)=3f(x)$$

$f(x)=ke^{3x}$

Since $f(0)=1, \quad \Rightarrow \quad k=1$

$$\Rightarrow \quad f(x)=e^{3x}$$

$$f'(x)=3e^{3x}$$

$$\Rightarrow \quad f'(15)=3e^{45}$$

18.(A) Point of intersection of $7x+2y-9=0$ and $x-7y+6=0$ is $(1, 1)$

Required equation of circle is $(x-2)^2+(y-1)^2=1^2$

19.(B) $\because \quad \frac{dy}{dx}+\sin\frac{x+y}{2}=\sin\frac{x-y}{2}$

$\Rightarrow \quad \frac{dy}{dx}=\sin\frac{x-y}{2}-\sin\frac{x+y}{2}$

$\Rightarrow \quad \frac{dy}{dx}=-2\cos\frac{x}{2}\cdot\sin\frac{y}{2}$

$\Rightarrow \quad \int \operatorname{cosec}\frac{y}{2}\,dy=-2\int\cos\frac{x}{2}\,dx$

$\Rightarrow \quad 2\log\tan\frac{y}{4}=-2\cdot 2\sin\frac{x}{2}+c'$

$\therefore \quad \log\left(\tan\frac{y}{4}\right)=c-2\sin\frac{x}{2}$

20.(A) $|z-1|+|z-4\sin\theta|=\sqrt{5}\,r$

For ellipse, $|4\sin\theta-1|<\sqrt{5}$

$\Rightarrow \quad \frac{1-\sqrt{5}}{4}<\sin\theta<\frac{1+\sqrt{5}}{4}$

$\therefore \quad -\sin\frac{\pi}{10}<\sin\theta<\sin\frac{3\pi}{10}$

SECTION-2

1.(3) $3^{\left(\frac{1}{4}+2\left(\frac{1}{8}\right)+3\left(\frac{1}{16}\right)+\ldots\infty\right)}=?$

Let $x=\frac{1}{4}+2\left(\frac{1}{8}\right)+3\left(\frac{1}{16}\right)+4\left(\frac{1}{32}\right)+\ldots+\infty\ (A.G.P.)$

$\frac{1}{2}\cdot x=\frac{1}{8}+2\left(\frac{1}{16}\right)+3\left(\frac{1}{32}\right)+\ldots\infty$

$\Rightarrow \quad x-\frac{x}{2}=\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\ldots\infty$

$\Rightarrow \quad \frac{x}{2}=\frac{\frac{1}{4}}{1-\frac{1}{2}}$

$\Rightarrow \quad \frac{x}{2}=\frac{1}{4}\times\frac{2}{1}=\frac{1}{2} \qquad \Rightarrow \quad x=1$

$\therefore \quad 3^{\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\frac{4}{32}+\ldots\infty}=3$

2.(6) $x^2+x+1=0$

$x=\omega, \omega^2$

Now, $x+\frac{1}{x}=\omega+\omega^2=-1$

$x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2=1-2=-1$

13.(B) $I=\int_{-1}^{1}\frac{dx}{(1+e^{x})(1+x^{2})}$ ....(1)

$I=\int_{-1}^{1}\frac{dx}{(1+e^{-x})(1+x^{2})}$ ...(2)

$2I=\int_{-1}^{1}\frac{1}{1+x^{2}}\left(\frac{1}{1+e^{x}}+\frac{1}{1+e^{-x}}\right)$

$2I=\int_{-1}^{1}\frac{dx}{1+x^{2}}$

$2I=2\int_{0}^{1}\frac{dx}{1+x^{2}}$

$I=(\tan^{-1}x)_{0}^{1}=\frac{\pi}{4}$

14.(C) We have $|A|=\begin{vmatrix}x+\lambda & x & x\\ x & x+\lambda & x\\ x & x & x+\lambda\end{vmatrix}$ $[C_1\to C_1+C_2+C_3]$

$=\begin{vmatrix}3x+\lambda & x & x\\ 3x+\lambda & x+\lambda & x\\ 3x+\lambda & x & x+\lambda\end{vmatrix}=(3x+\lambda)\begin{vmatrix}1 & x & x\\ 1 & x+\lambda & x\\ 1 & x & x+\lambda\end{vmatrix}$

$=(3x+\lambda)\begin{vmatrix}1 & x & x\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{vmatrix}=\lambda^{2}(3x+\lambda)$

[Take $3x+\lambda$ common and use $R_2\to R_2-R_1, R_3\to R_3-R_1$]

Thus, A^{-1} will exist if $\lambda\neq 0$ and $3x+\lambda\neq 0$.

15.(B) As $f(\alpha)=f(\beta)=f(\gamma)=0$

$\Rightarrow f(x)$ is not one-one.

If $y=(x-\alpha)(x-\beta)(x-\gamma)$

Any odd degree polynomial is always onto

16.(C) $\because$ $|\vec{a}+2\vec{b}+3\vec{c}|=\sqrt{3+2\sqrt{2}}$

$\therefore$ $4\cos\alpha+6\cos\beta=-11+2\sqrt{2}-12\cos\theta$

The value of $4\cos\alpha+6\cos\beta$ will be maximum when $\theta=\frac{2\pi}{3}$ i.e., $\cos\theta=-\frac{1}{2}$

$\therefore$ The maximum value of $4\cos\alpha+6\cos\beta$ is $2\sqrt{2}-5$

17.(B) Let $(2x^{2}-3x+1)^{11}=a_0+a_1x+a_2x^{2}+...+a_{22}x^{22}$;

To find $a_0+a_2+a_4+...+a_{22}$?

$S_E+S_O=P(1)=0$ where $P(x)=(2x^{2}-3x+1)^{11}$ ...(1)

$S_E-S_O=P(-1)=6^{11}$ ....(2)

$\Rightarrow$ $2S_E=6^{11}$ $\Rightarrow$ $S_E=3\cdot 6^{10}$

6.(2) Given $\vec{c}=(2\vec{a}\times\vec{b})-3\vec{b}$

$\vec{b}\cdot\vec{c}=\vec{b}\cdot\{(2\vec{a}\times\vec{b})-3\vec{b}\}$

$\vec{b}\cdot\vec{c}=-3\,|\vec{b}\,|^2$ …(i)

Now $|\vec{a}\times\vec{b}\,|^2=|\vec{a}\,|^2\,|\vec{b}\,|^2-(\vec{a}\cdot\vec{b})^2$

$=16-4=12$

And $|\vec{c}\,|^2=|(2\vec{a}\times\vec{b})-3\vec{b}\,|^2$

$|\vec{c}\,|^2=4\,|\vec{a}\times\vec{b}\,|^2+9|\vec{b}|^2-\underbrace{6\vec{b}.(2\vec{a}\times\vec{b})}_{zero}$

$= 48+ 144 = 192$

$|\vec{c}\,|=8\sqrt{3}$

Now, $\cos\theta=\dfrac{\vec{b}\cdot\vec{c}}{|\vec{b}\,||\vec{c}\,|}=\dfrac{-3\,|\vec{b}\,|^2}{|\vec{b}\,||\vec{c}\,|}=\dfrac{-3\,|\vec{b}\,|}{|\vec{c}\,|}=\dfrac{-3(4)}{8\sqrt{3}}$

$=\dfrac{-\sqrt{3}}{2}$

Hence $\theta=\dfrac{5\pi}{6}$ $\quad\therefore[\theta]=2$

7.(1) If $e^{-\int\frac{x}{1+x}dx}=e^{-\int\frac{x+1-1}{x+1}dx}=e^{-x+\ell n\,|x+1|}=e^{-x}\cdot(x+1)$

$y\cdot(x+1)e^{-x}=\int\dfrac{x+1}{x+1}e^{-x}dx+c$

$y\cdot(x+1)\,e^{-x}=-e^{-x}+c$

$\Rightarrow\quad y=-\dfrac{1}{x+1}+\dfrac{ce^x}{x+1}$

$x=0, y=-1\ \Rightarrow\ -1=-1+c\ \Rightarrow\ c=0$

$\Rightarrow\quad y=-\dfrac{1}{1+x}$

$y(2)=-\dfrac{1}{3}$

8.(0) We have $g(x)=f\left(\dfrac{x}{f(x)}\right)$

On differentiating w.r.t. x we get

$g'(x)=f'\left(\dfrac{x}{f(x)}\right)\times\left(\dfrac{f(x)-xf'(x)}{f^2(x)}\right)$

$\therefore\quad f'(1)=f'\left(\dfrac{1}{f(1)}\right)\times\left(\dfrac{f(1)-f'(1)}{f^2(1)}\right)$

As $f(1)=f'(1)$

$\Rightarrow\quad g'(1)=0$

9.(4) $B=A\left(I+4A+6A^2+4A^3+A^4\right)=A(I+A)^4$

$|B|=|A|\,|I+A|^4=1$ (Given) (As $|A|=1$)

$\Rightarrow \quad \begin{vmatrix} 1+\cos\alpha & \sin\alpha \\ -\sin\alpha & 1+\cos\alpha \end{vmatrix}^4 = 1 \quad \Rightarrow \quad [(1+\cos\alpha)^2 + \sin^2\alpha]^4 = 1$

$\Rightarrow \quad 16(1+\cos\alpha)^4 = 1 \quad \Rightarrow \quad 1+\cos\alpha = \frac{1}{2}$

$\Rightarrow \quad \cos\alpha = -\frac{1}{2}$

$\therefore$ Number of values of α in $[-2\pi, 2\pi]$ are 4.

10.(32) Let $\frac{\pi}{31} = \theta$, then

$$\cos\theta\cos 2\theta\cos 4\theta\cos 8\theta\cos 16\theta = \frac{(2\sin\theta\sin\theta)\cos 2\theta\cos 4\theta\cos 8\theta\cos 16\theta}{2\sin\theta}$$

$$= \frac{(2\sin 2\theta\cos 2\theta)\cos 4\theta\cos 8\theta\cos 16\theta}{4\sin\theta} = \frac{(2\sin 4\theta\cos 4\theta)\cos 8\theta\cos 16\theta}{8\sin\theta}$$

$$= \frac{(2\sin 8\theta\cos 8\theta)\cos 16\theta}{16\sin\theta} = \frac{2\sin 16\theta\cos 16\theta}{32\sin\theta} = \frac{\sin 32\theta}{32\sin\theta} = \frac{\sin(\pi+\theta)}{32\sin\theta} = \frac{-1}{32}$$


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